

MATH 6241 – Stochastic Processes

❖ Section 9. Poisson Process

Instructor: He Wang
Department of Mathematics
Northeastern University

Counting Process

Poisson process-counting

Poisson process-time

Thinning

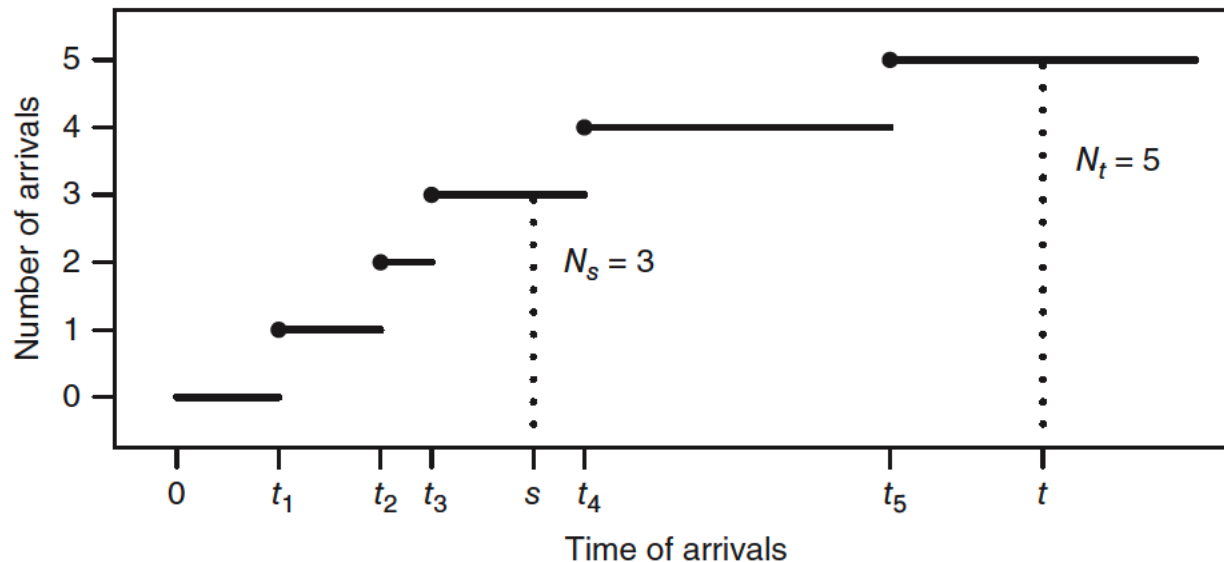
Superposition

Counting Process

A stochastic process $(N_t)_{t \geq 0}$ is a **counting process** if N_t is the number of events occurred from time 0 up to and including time t . We assume

- $N_t \in \{0, 1, 2, \dots\}$ for all t ,
- $N_0 = 0$,
- If $0 \leq s \leq t$, then $N_s \leq N_t$.

So $N(t) := N_t$ counts how many events have occurred by time t .



Note:

- If $0 \leq s < t$, then $N_t - N_s$ is the **number of events occurred during the interval $(s, t]$** . This is the fundamental object for probability calculations.
- A realization of N_t looks like a non-decreasing right-continuous step function: Jumps upward by exactly 1 at each arrival time.
- Methods of counting processes include **Poisson processes** and **Renewal processes**.
- Real world problems include texts arriving, customers entering a store, radioactive decays, goals in hockey, babies born, etc.

Poisson random variable

Recall that a discrete random variable $X \sim \text{Poisson}(\lambda)$ is said to be a Poisson random variable with parameter λ

$$\mathbb{P}(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

for $k = 0, 1, 2, 3, \dots$

Properties:

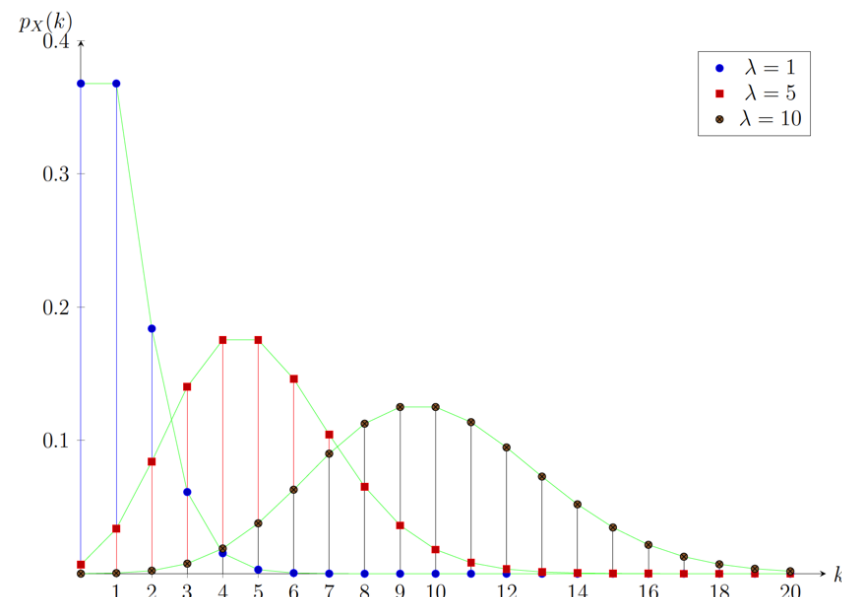
- If $X \sim \text{Poisson}(\lambda)$, then $E[X] = \lambda$ and $\text{Var}(X) = \lambda$
- If $X_i \sim \text{Poisson}(\lambda_i)$ and independent, then

$$X_1 + \dots + X_n \sim \text{Poisson}(\lambda_1 + \dots + \lambda_n)$$

- The Poisson distribution can be viewed as the limit of binomial distribution.

$$Y_n \sim \text{Binomial}(n, p_n)$$

$$\lim_{n \rightarrow \infty} np_n = \lambda$$



Poisson process (Definition 1: Counting View)

Definition: A **Poisson process with rate** $\lambda > 0$ is a counting process $(N_t)_{t \geq 0}$ such that:

(1) Initialization: $N_0 = 0$.

(2) Marginal distributions. For every $t > 0$,

$$N_t \sim \text{Poisson}(\lambda t) \quad \mathbb{P}(N_t = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

(3) Stationary increments

$$\text{For all } s, t > 0, \quad N_{t+s} - N_s \stackrel{d}{=} N_t.$$

(4) Independent increments

$$N_t - N_s \text{ is independent of } N_r - N_q \text{ (} 0 \leq q < r \leq s < t \text{)}.$$

A Poisson process describes random events occurring independently, at a constant average rate, with no memory.

Remarks:

Poisson marginal distributions imply $E[N_t] = \lambda t$.

Stationary increments: Only the **length** of the interval matters, not its location.

Independent increments: Counts on disjoint time intervals do not influence each other.

Warning: $N_{t+s} - N_s \stackrel{d}{=} N_t$ only means they have the same distribution.

But they are **not** the same random variable.

For example, $N_7 - N_2 \neq N_5$ as random variables.

And N_5 is **not independent** of N_2

Example: Bob receives text messages starting at 9 a.m., according to a Poisson process with rate

$$\lambda = 8 \text{ texts/hour}$$

Find the probability that he will receive exactly 15 texts by 11 a.m., and exactly 55 texts by 4 p.m.

$$\text{Find } \mathbb{P}(N_2 = 15, N_7 = 55)$$

$$\{N_2 = 15, N_7 = 55\} = \{N_2 = 15, N_7 - N_2 = 40\}$$

N_2, N_7 are dependent, however, N_2 and $N_7 - N_2$ are independent

$$\mathbb{P}(N_2 = 15, N_7 = 55) = \mathbb{P}(N_2 = 15, N_7 - N_2 = 40) = \mathbb{P}(N_2 = 15)\mathbb{P}(N_7 - N_2 = 40)$$

$$N_2 \sim \text{Poisson}(8 \times 2) = \text{Poisson}(16)$$

$$N_7 - N_2 \sim \text{Poisson}(8 \times 5) = \text{Poisson}(40)$$

$$\mathbb{P}(N_2 = 15) = \frac{e^{-16} 16^{15}}{15!}, \quad \mathbb{P}(N_7 - N_2 = 40) = \frac{e^{-40} 40^{40}}{40!}.$$

Translating Time (Shift Invariance)

Let $(N_t)_{t \geq 0}$ be Poisson process with rate λ .

Define the **translated process**:

$$\tilde{N}_t := N_{t+s} - N_s, \quad t \geq 0.$$

Then, $(\tilde{N}_t)_{t \geq 0}$ be Poisson process with rate λ .

Once you condition on what happened up to time s , the future behaves like a brand-new Poisson process.

Example. At a coffee shop, customers arrive according to a Poisson process with rate

$$\lambda = 60 \text{ customers/hour.}$$

If 85 customers arrive during the first hour, what is the probability that at most 190 customers arrive during the first three hours?

Let N_t be the number of arrivals in the first t hours.

$$\begin{aligned}\mathbb{P}(N_3 \leq 190 | N_1 = 85) &= \mathbb{P}(N_3 - N_1 \leq 105 | N_1 = 85) \\ &= \mathbb{P}(N_3 - N_1 \leq 105 | N_1 = 85) \\ &= \mathbb{P}(N_2 \leq 105) \\ &= \sum_{k=0}^{105} e^{-120} \frac{120^k}{k!}\end{aligned}$$

Here, $N_3 - N_1 \sim \text{Poisson}(60 \cdot 2) = \text{Poisson}(120)$.

□ Arrival and Interarrival Times

Let

X_k := **interarrival time** between the $(k - 1)$ -st and k -th arrivals,

$S_n := X_1 + \cdots + X_n$ = **arrival time** of the n -th arrival.

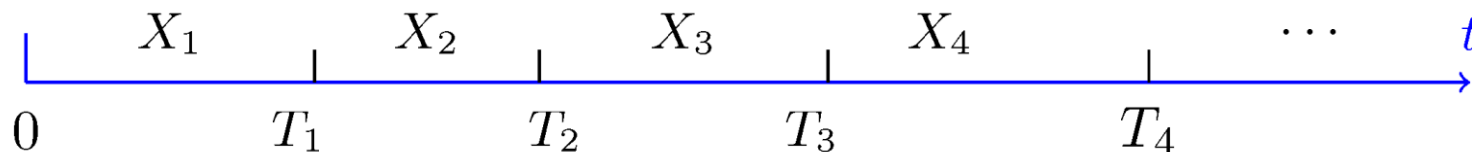
First arrival

$$\mathbb{P}(X_1 > t) = P(N_t = 0) = e^{-\lambda t}$$

So,

$X_1 \sim \text{Exponential}(\lambda)$.

PDF $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$



Let $s > 0$ and $t > 0$.

The two intervals $(0, s]$ and $(s, s + t]$ are disjoint. We can write

$$\begin{aligned}\mathbb{P}(X_2 \geq t | X_1 = s) &= \mathbb{P}(\text{no arrival in } (s, s + t] | X_1 = s) \\ &= \mathbb{P}(\text{no arrival in } (s, s + t]) \\ &= e^{-\lambda t}\end{aligned}$$

Moreover, $X_1, X_2, X_3, \dots$ are i.i.d. *Exponential*(λ).

Poisson process is exactly a counting process with i.i.d. exponential interarrival times.

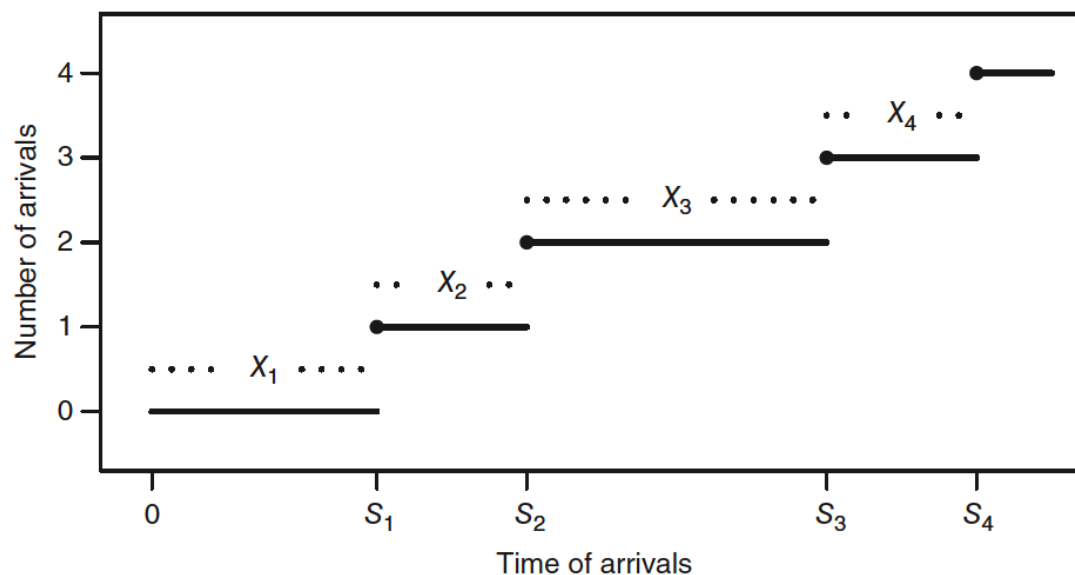
Poisson Process-Definition 2

Let $X_1, X_2, \dots$ be i.i.d. exponential random variables $X_i \sim \text{Exponential}(\lambda)$.

Let $N_0 = 0$, and for $t > 0$,

$$N_t = \max\{n: X_1 + \dots + X_n \leq t\}$$

Then, $(N_t)_{t \geq 0}$ is a Poisson process with rate λ .



Memoryless Property

A nonnegative random variable X is **memoryless** if for all $s, t > 0$,

$$P(X \geq t + s \mid X > s) = P(X \geq t)$$

The exponential is the only continuous memoryless distribution

$$P(X \geq t + s \mid X > s) = \frac{P(X \geq t + s, X > s)}{P(X > s)}$$

$$= \frac{P(X \geq t + s)}{P(X > s)}$$

$$= \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}}$$

$$= e^{-\lambda t} = P(X \geq t)$$

Bus-stop story (Amy vs Zach)

Assume bus arrivals form a Poisson process with rate $\lambda = 1/30$ per minute, so the waiting time for the next bus is

$$A \sim \text{Exp}(\lambda), \text{ and } P(A > t) = e^{-\lambda t} = e^{-t/30}.$$

Amy arrives at time 0. Zach arrives at time 10.

Let Z be Zach's waiting time after he arrives.

Claim: $Z \sim \text{Exp}(\lambda)$ and hence $E[Z] = 30$ minutes (same as Amy).

Zach waits more than t minutes iff **(i)** no bus arrives in the first 10 minutes after Amy, and **(ii)** Amy's waiting time exceeds $10 + t$. Thus

$$P(Z > t) = P(A > 10 + t \mid A > 10).$$

$$= P(A > t)$$

$$= e^{-t/30}$$

In the bus-stop story, we assume:

- Bus arrivals follow a **Poisson process**
- Interarrival times are **independent**
- Each waiting time is **exponentially distributed**

This implies a very strong assumption:

- **The bus has no clock.**

Question: If bus arrivals were **uniformly distributed** over $[0,60]$, would memorylessness hold?

Minimum of Independent Exponentials

Theorem. Let $X_1, \dots, X_n$ be independent with $X_k \sim \text{Exp}(\lambda_k)$. Define

$$M = \min(X_1, \dots, X_n).$$

Then, $M \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$, i.e.,

$$P(M > t) = e^{-t(\lambda_1 + \dots + \lambda_n)}.$$

For each k ,

$$P(M = X_k) = \frac{\lambda_k}{\lambda_1 + \dots + \lambda_n}$$

Example A Boston subway station services the red, green, and orange lines. Subways on each line arrive at the station according to three independent Poisson processes. On average, there is one red train every 10 minutes, one green train every 15 minutes, and one orange train every 20 minutes.

- (i) When you arrive at the station what is the probability that the first subway that arrives is for the green line?
- (ii) How long will you wait, on average, before some train arrives?
- (iii) You have been waiting 20 minutes for a red train and have watched three orange trains go by. What is the expected additional time you will wait for your subway?

Example (Boston station: red/green/orange)

Rates:

- Red: mean 10min $\Rightarrow \lambda_R = 1/10$
- Green: mean 15min $\Rightarrow \lambda_G = 1/15$
- Orange: mean 20min $\Rightarrow \lambda_O = 1/20$

Let X_R, X_G, X_O be independent exponentials with those rates.

(i) Probability the first arriving train is green.

By the “who is the minimum” formula:

$$\mathbb{P}(\text{first is green}) = \mathbb{P}(\min(X_R, X_G, X_O) = X_G) = \frac{\lambda_G}{\lambda_R + \lambda_G + \lambda_O} = \dots = \frac{4}{13}$$

(ii) Average waiting time until *some* train arrives

The minimum

$$M = \min(X_R, X_G, X_O) \sim \text{Exp}\left(\frac{13}{60}\right)$$

So,

$$\mathbb{E}[M] = \frac{1}{13/60} = \frac{60}{13} \approx 4.615 \text{ minutes.}$$

(iii) You've waited 20 minutes for red and saw 3 orange trains; expected additional time to wait for red.

Key point: the red process is independent of the orange (and green) processes, and the red interarrival time is exponential (memoryless).

Conditioning on “no red yet by 20”,

$$E[X_R - 20 \mid X_R > 20] = E[X_R] = 10 \text{ minutes.}$$

❑ Arrival Times Distribution

Let S_n be the arrival time of the n -th arrival in a Poisson process of rate λ . Then

$$S_n := X_1 + \cdots + X_n \text{ and } X_i \text{ are iid and } \sim \text{Exp}(\lambda)$$

So,.

$$S_n \sim \text{Gamma}(n, \lambda) \text{ (called Erlang when } n \in \mathbb{N})$$

The PDF is

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \quad t > 0,$$

The mean and variance are:

$$\mathbb{E}[S_n] = \frac{n}{\lambda}, \quad \text{Var}(S_n) = \frac{n}{\lambda^2}.$$

Example (NBA): Scoring as a Poisson Process

A particular NBA player's **made 3-point shots** are modeled as a Poisson process. Assume the **average time between made 3s is 10 minutes**.

$$\lambda = \frac{1}{10} \text{ (per minute)}$$

Let S_n be the time (in minutes) of the n -th made 3-pointer. Then

$$S_n \sim \text{Gamma}(n, \lambda)$$

(i) In a 48-minute game, find the probability that the **4th made 3** occurs in the **last 4 minutes** of the game.

$$\mathbb{P}(44 < S_4 \leq 48) = \int_{44}^{48} \frac{\lambda^4 t^3 e^{-\lambda t}}{3!} dt = \frac{1}{6} \int_{44}^{48} \left(\frac{1}{10}\right)^4 t^3 e^{-t/10} dt.$$

$$\approx 0.0652$$

(ii) Mean time of the 3rd made 3, given at least 3 are made

$$\mathbb{E}(S_3 \mid S_3 < 48) = \frac{1}{\mathbb{P}(S_3 < 48)} \int_0^{48} t f_{S_3}(t) dt.$$

$S_3 \sim \text{Gamma}(3, \lambda)$ and pdf

$$f_{S_3}(t) = \frac{\lambda^3 t^2 e^{-\lambda t}}{2}.$$

$$\mathbb{E}(S_3 \mid S_3 < 48) = \frac{1}{\mathbb{P}(S_3 < 48)} \int_0^{48} t \cdot \frac{(1/10)^3 t^2 e^{-t/10}}{2} dt.$$

$$\approx 24.693 \text{ minutes}$$

```
import numpy as np
from scipy.stats import gamma
from scipy.integrate import quad

# parameters
lam = 1/10
shape = 4
t1, t2 = 44, 48
prob = gamma.cdf(t2, a=shape, scale=1/lam) - gamma.cdf(t1, a=shape, scale=1/lam)
prob
```

```
# parameters
shape = 3
# denominator:  $P(S_3 < 48)$ 
den = gamma.cdf(48, a=shape, scale=1/lam)

# numerator:  $E[S_3 * 1_{\{S_3 < 48\}}]$ 
num, _ = quad(lambda t: t * gamma.pdf(t, a=shape, scale=1/lam), 0, 48)

num, den, num / den
```

Monte Carlo Check

```
rng = np.random.default_rng(0)

# simulate many games
N = 200_000
S3 = rng.gamma(shape=3, scale=1/lam, size=N)
S4 = rng.gamma(shape=4, scale=1/lam, size=N)

# part (i)
mc_prob = np.mean((S4 > 44) & (S4 <= 48))

# part (ii)
mc_mean = np.mean(S3[S3 < 48])

mc_prob, mc_mean

(0.0649, 24.70)
```

Thinning a Poisson Process

Let $(N_t)_{t \geq 0}$ be a Poisson process with rate λ .

For each arrival, independently assign a mark/type in $\{1, 2, \dots, n\}$ with probability p_k

$$p_1 + \dots + p_n = 1$$

Let $N_t^{(k)}$ be the number of arrivals in $[0, t]$ who mark equal k .

Theorem:

1. For each k , $\{N_t^{(k)}\}_{t \geq 0}$ is a Poisson process with rate λp_k
2. The processes $\{N_t^{(k)}\}_{t \geq 0}$ are independent.
3. For each t ,
$$N_t = N_t^{(1)} + \dots + N_t^{(n)}$$

Example (Customer calls)

Calls arrive to a help desk as a Poisson process with rate $\lambda = 12$ per hour.

Each call is a **billing** call with probability $p = 0.35$ (otherwise it's **technical**), independently.

Let B_t # = billing calls in t hours, T_t # = technical calls.

(1) In 30 minutes, what is the expectation and standard deviation of the number of billing calls?

Billing calls B_t form a thinned Poisson process with rate $\lambda_B = \lambda p = 12(0.35) = 4.2$ per hour.

$$B_{0.5} \sim \text{Poisson}(4.2 \times 0.5) = \text{Poisson}(2.1).$$

Hence

$$E[B_{0.5}] = 2.1, \quad SD(B_{0.5}) = \sqrt{2.1} \approx 1.449.$$

(ii) In the next 10 minutes, find $P(\text{no billing calls but at least one technical call})$.

$t = 10 \text{ minutes} = 1/6 \text{ hour}$. We want:

$$P(B_t = 0, T_t > 0) = P(B_t = 0)P(T_t > 0).$$

Rates: $\lambda_B = 4.2, \lambda_T = 12(0.65) = 7.8$.

$$B_t \sim \text{Poisson}\left(\frac{4.2}{6}\right) = \text{Poisson}(0.7), T_t \sim \text{Poisson}\left(\frac{7.8}{6}\right) = \text{Poisson}(1.3).$$

$$P(B_t = 0, T_t > 0) = P(B_t = 0)P(T_t > 0) = e^{-0.7} (1 - e^{-1.3}) \approx 0.361$$

(iii) Given that 9 calls arrived in an hour, find $P(\text{exactly 4 are billing})$.

$$B \mid (N = 9) \sim \text{Binomial}(9, 0.35),$$

$$P(B = 4 \mid N = 9) = \binom{9}{4} (0.35)^4 (0.65)^5 \approx 0.266.$$

Proof of the theorem in the special case:

For $b, t \in \{0, 1, 2, \dots\}$,

$$\begin{aligned}\mathbb{P}(B_t = b, T_t = t) &= \mathbb{P}(B_t = b, T_t = t, N_t = b + t) \\ &= \mathbb{P}(B_t = b, T_t = t \mid N_t = b + t) \mathbb{P}(N_t = b + t).\end{aligned}$$

Given $N_t = b + t$, the $b + t$ marks are i.i.d., so

$$\mathbb{P}(B_t = b, T_t = t \mid N_t = b + t) = \binom{b+t}{b} p^b (1-p)^t = \frac{(b+t)!}{b! t!} p^b (1-p)^t.$$

Also, since $N_t \sim \text{Poisson}(\lambda t)$,

$$\mathbb{P}(N_t = b + t) = e^{-\lambda t} \frac{(\lambda t)^{b+t}}{(b+t)!}.$$

Multiply:

$$\begin{aligned}\mathbb{P}(B_t = b, T_t = t) &= \frac{(b+t)!}{b! t!} p^b (1-p)^t \cdot e^{-\lambda t} \frac{(\lambda t)^{b+t}}{(b+t)!} \\ &= \frac{e^{-\lambda t} (\lambda p t)^b (\lambda (1-p) t)^t}{b! t!} \\ &= \left(\frac{e^{-\lambda p t} (\lambda p t)^b}{b!} \right) \left(\frac{e^{-\lambda (1-p) t} (\lambda (1-p) t)^t}{t!} \right),\end{aligned}$$

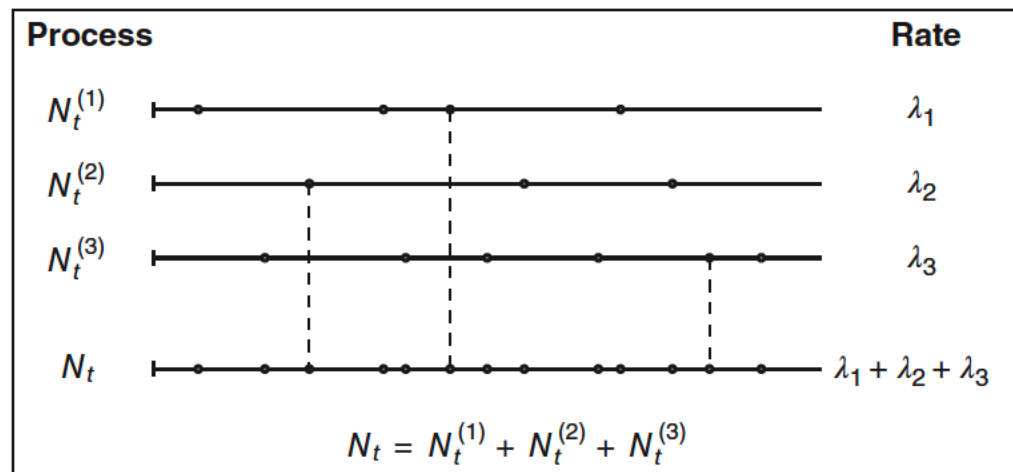
using $e^{-\lambda t} = e^{-\lambda p t} e^{-\lambda (1-p) t}$.

Superposition Process

Let $\{N_t^{(1)}\}_{t \geq 0}, \dots, \{N_t^{(n)}\}_{t \geq 0}$ be independent Poisson processes with rates $\lambda_1, \dots, \lambda_n$.

Define
$$N_t = N_t^{(1)} + \dots + N_t^{(n)}$$

Then, $\{N_t\}_{t \geq 0}$ is a Poisson processes with rate $\lambda = \lambda_1 + \dots + \lambda_n$



Example. In a certain city, **cars**, **trucks**, and **motorcycles** pass through an intersection according to **independent Poisson processes** with respective rates

$$\lambda_C, \lambda_T, \lambda_M \text{ (vehicles per hour).}$$

Let C_t , T_t , M_t denote the numbers of cars, trucks, and motorcycles observed by time t , and let

$$N_t = C_t + T_t + M_t$$

be the total number of vehicles.

(i) Probability of seeing no vehicles in the first 2 hours

Since N_t is the superposition of three independent Poisson processes,

$$N_t \sim \text{Poisson}((\lambda_C + \lambda_T + \lambda_M)t).$$

$$P(N_2 = 0) = e^{-2(\lambda_C + \lambda_T + \lambda_M)}.$$

(ii) Given that 4 vehicles were observed in 1 hour, find the probability that exactly one was of each type except one extra car.

$$P(C_1 = 2, T_1 = 1, M_1 = 1 | N_1 = 4)$$

Conditioned on $N_1 = 4$, the vector (C_1, T_1, M_1) has a **multinomial distribution** with parameters

$$n = 4, \quad \left(\frac{\lambda_C}{\lambda_C + \lambda_T + \lambda_M}, \frac{\lambda_T}{\lambda_C + \lambda_T + \lambda_M}, \frac{\lambda_M}{\lambda_C + \lambda_T + \lambda_M} \right)$$

$$P(C_1 = 2, T_1 = 1, M_1 = 1 | N_1 = 4)$$

$$= \frac{4!}{2! 1! 1!} \left(\frac{\lambda_C}{\lambda_C + \lambda_T + \lambda_M} \right)^2 \left(\frac{\lambda_T}{\lambda_C + \lambda_T + \lambda_M} \right) \left(\frac{\lambda_M}{\lambda_C + \lambda_T + \lambda_M} \right).$$

Method2.

$$\begin{aligned} P(C_1 = 2, T_1 = 1, M_1 = 1) &= P(C_1 = 2)P(T_1 = 1)P(M_1 = 1) \\ &= \left(e^{-\lambda_C} \frac{\lambda_C^2}{2!} \right) (e^{-\lambda_T} \lambda_T) (e^{-\lambda_M} \lambda_M) . \end{aligned}$$

$$P(N_1 = 4) = e^{-(\lambda_C + \lambda_T + \lambda_M)} \frac{(\lambda_C + \lambda_T + \lambda_M)^4}{4!}$$

So

$$P(C_1 = 2, T_1 = 1, M_1 = 1 \mid N_1 = 4) = \frac{P(C_1 = 2, T_1 = 1, M_1 = 1)}{P(N_1 = 4)}$$

Embedding and the Birthday Problem

Discrete probability problems can often be analyzed more effectively by **embedding** them into continuous-time stochastic models.

1. The classical birthday problem (discrete formulation)

Assume that each person's birthday is independently and uniformly distributed over the 365 days of the year (ignoring leap years).

How many people must be in a room before the probability that at least two people share a birthday is at least 50%?

Let p_k denote the probability that **at least one shared birthday** occurs among k people. Then

$$\begin{aligned} p_k &= 1 - P(\text{all birthdays are distinct}). \\ &= 1 - \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365 - (k - 1)}{365} \end{aligned}$$

$$p_{22} \approx 0.476, p_{23} \approx 0.507.$$

2 A different question: when does the first match occur?

*As people enter the room one by one, how many people are present at the **first** moment when two people share a birthday?*

Let K denote this random number of people. Our goal is to compute $E[K]$ and $\text{Var}(K)$.

To do this, we embed the discrete arrival process into continuous time.

Assume that people enter the room according to a **Poisson process**

$$(N_t)_{t \geq 0}$$

with rate $\lambda = 1$ (one person per unit time on average).

Each arriving person is independently **marked** with one of the 365 birthdays, each with probability $1/365$.

365 independent **thinned Poisson processes**

$$(N_t^{(1)}), (N_t^{(2)}), \dots, (N_t^{(365)}),$$

The superposition satisfies

$$N_t = N_t^{(1)} + \dots + N_t^{(365)}.$$

Linking continuous time T and discrete counts K

Let $X_1, X_2, \dots$ denote the interarrival times of the Poisson process N_t .
Then the X_i are i.i.d. exponential random variables with mean 1.

Let T be the **first time** at which two people in the room share the same birthday.

Since the K -th arrival occurs at time

$$T = \sum_{i=1}^K X_i,$$

$$E[T] = E[K]E[X_1] = E[K].$$

Arrival Times and Uniform Distribution

Let $(N_t)_{t \geq 0}$ be a Poisson process with rate λ , and let $S_1 < S_2 < \dots < S_n$ be the arrival times.

Conditional on $N_t = n$, the joint density of $(S_1, \dots, S_n)$ is

$$f(s_1, \dots, s_n) = \frac{n!}{t^n} \quad 0 < s_1 < \dots < s_n < t.$$

Equivalently,

$$(S_1, \dots, S_n) \stackrel{d}{=} (U_{(1)}, \dots, U_{(n)}).$$

U_i iid $Uniform(0, t)$

Poisson Process as a Markov Process

Continuous-time Markov chain

Birth Process View

State space: $\Omega = \{0, 1, 2, 3, \dots\}$

Transition rate:

$$q_{n,n+1} = \lambda$$

Generator

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & \dots \\ 0 & -\lambda & \lambda & \dots \\ \vdots & & & \end{pmatrix}$$

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross